(a)

Data Construction Pipeline

Textual Skeleton

G_{KG}

Subgraph Extraction

$G_T(s)$

Web Info Expansion

$G_T(s)$

$\mathcal{A}(v)$

$\mathcal{C}_T(s)$

Multimodal Pattern Expansion

(1) Text-Only

(2) Image-to-Text

Image Node

(3) Text-to-Image

Visual Attribute

Image Node

Exploration-Driven Selection

Multiple rollouts for each question

Exploration degree distribution across questions

Filter

Keep Top 30%

Curated Outputs

Cold-start tool-use trajectories (SFT)

8K

Exploration-rich QA pairs (RL)

10K

(b)

UniBrowse Agent Browsing

Multimodal Question

Identify the campus associated with this publication, then trace it to its parent university system. What animal appears on that university system‘s official seal?

Reverse Image Search

Query: image of the question

Result:

Philippine Collegian

The image matches the student publication Philippine Collegian.

Image to Text

Text Search

Query: Philippine Collegian campus

Result:

UP Diliman

Evidence indicates Philippine Collegian is associated with the UP Diliman campus.

Text Only

Image Search

Query: official seal of UP

Result:

Official seal of UP

The official seal is verified visually and its central animal is an eagle.

Text to Image

Final Answer

EAGLE